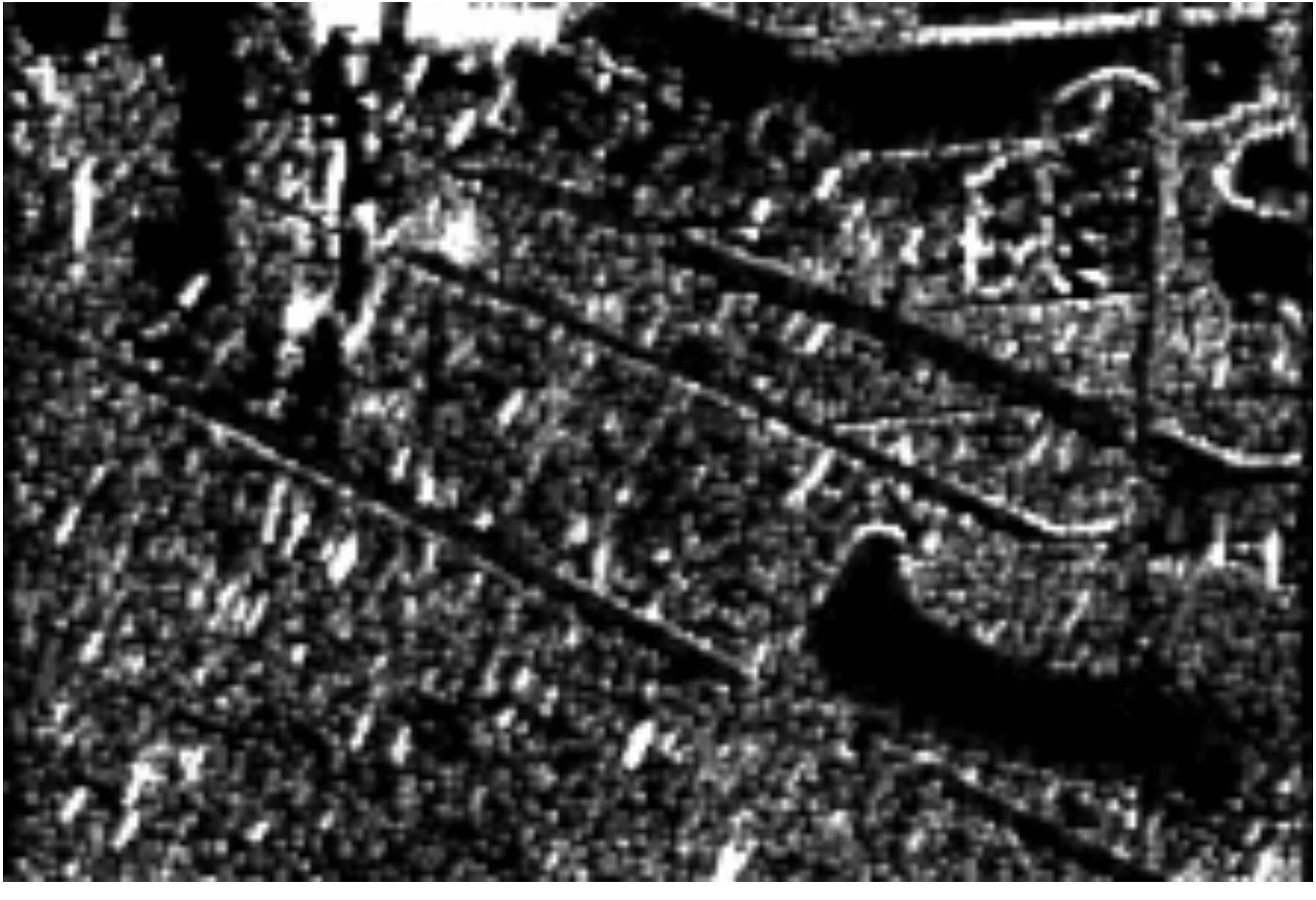
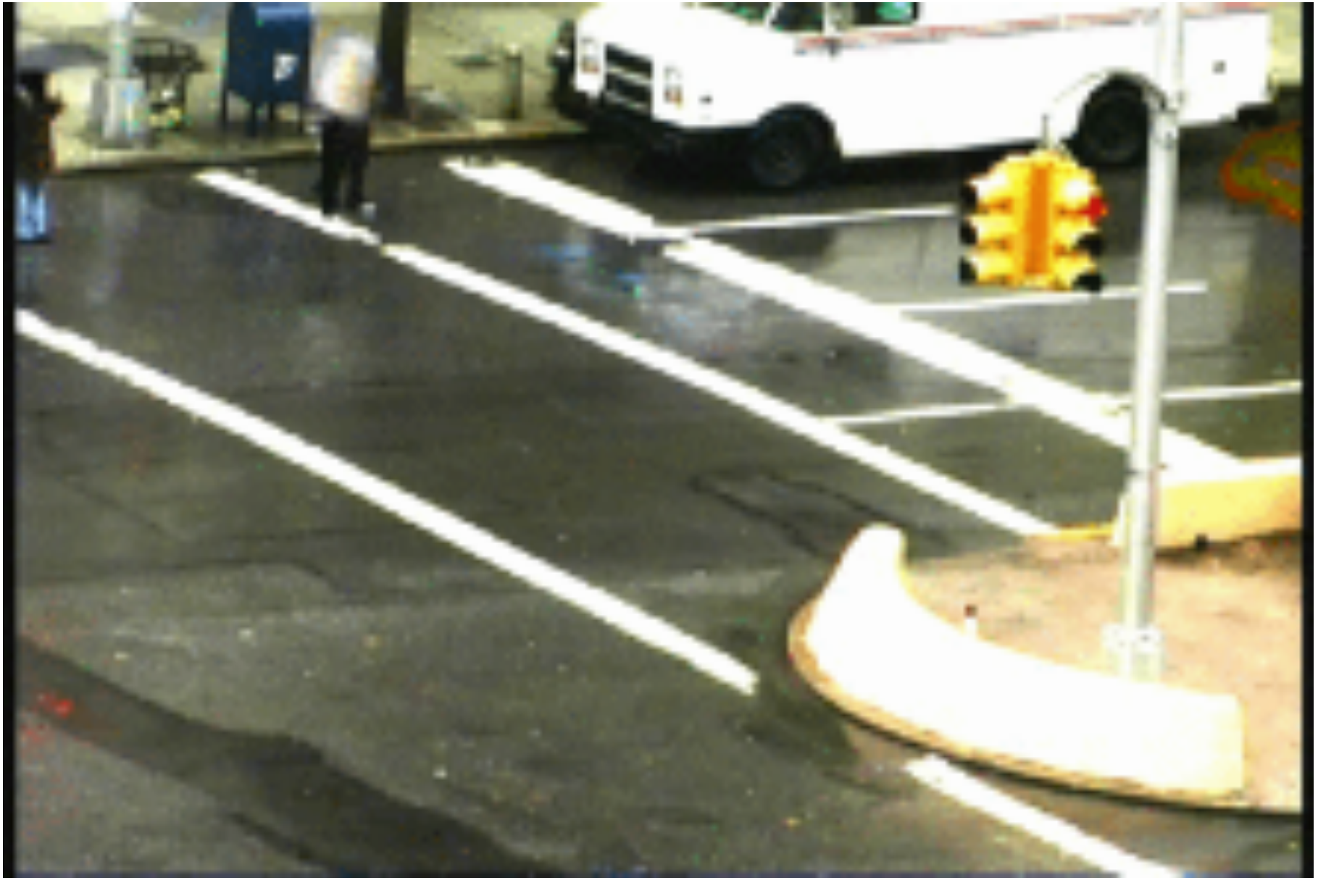
Background

Rain layer





Video

Video Decomposition

Moving objects (Foreground)

$$\mathcal{B} = \text{Fold}(UV^T),$$

$$U \in R^{d \times r}, V \in R^{n \times r},\ r < \min(d, n).$$

Low Rank Matrix Decomposition

Temporal redundancy

$$\mathcal{R} = \sum_{k=1}^{K} \sum_{s=1}^{n_k} D_{ks} \otimes \mathcal{M}_{ks}.$$

Sparse & repetitive patterns at multiple scales

Multi-scale Convolutional Sparse Coding

Markov Random Filed:

$$\mathcal{H}_{ijn} = \begin{cases} 1, & \text{location } ijn \text{ is moving objects,} \\ 0, & \text{location } ijn \text{ is background.} \end{cases}$$

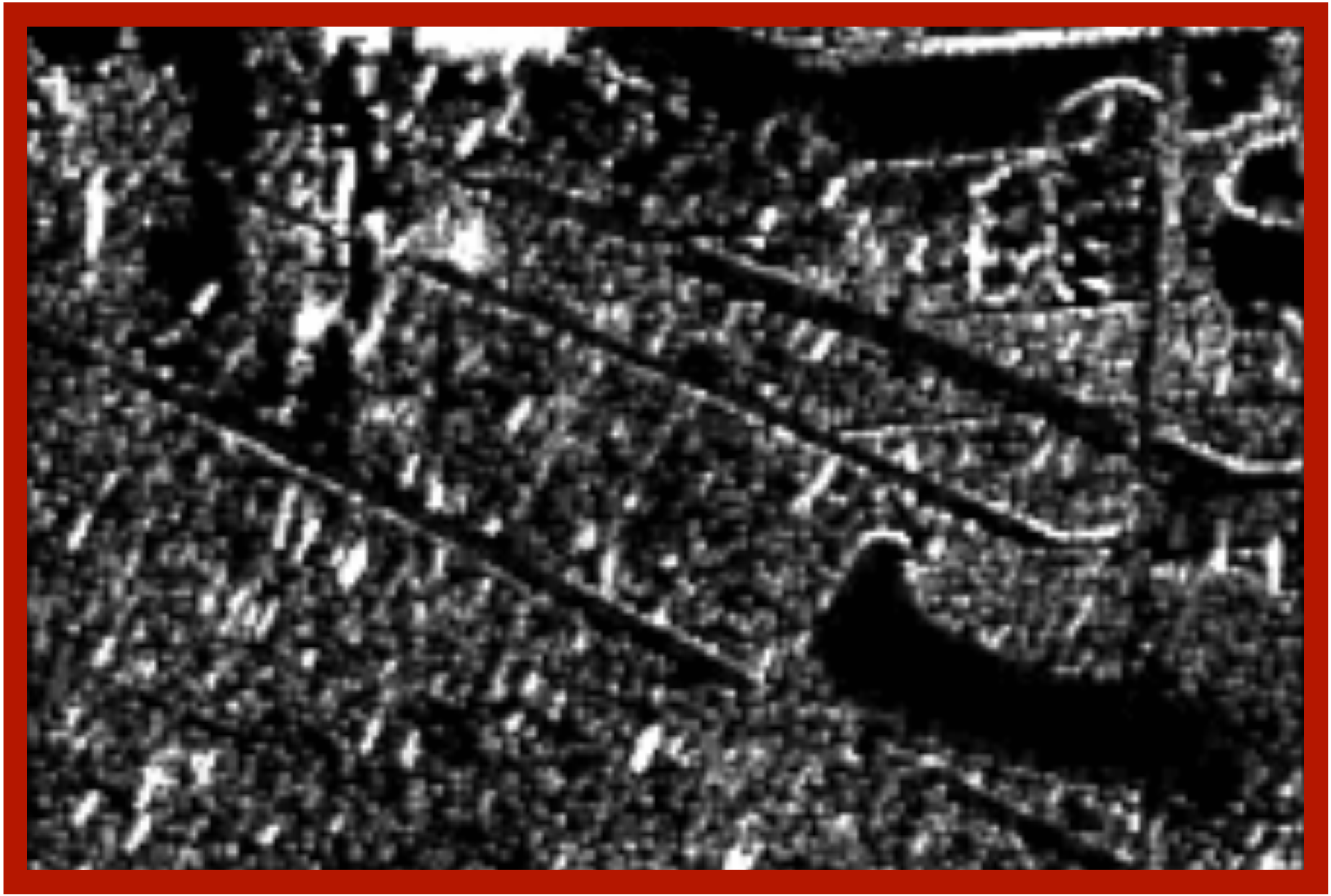
$$\mathcal{H} \circ \mathcal{X} = \mathcal{H} \circ (\mathcal{F} + \mathcal{R}).$$

Motion consistency over time

Each component follows certain physical or statistical properties!

1

OT-MSCSC: Online Rain/Snow Removal from Surveillance Videos (TIP 2021). **First author, high-impact, used in real-world environments**



(Foreground)

Moving objects

Temporal

redundancy

Sparse &

at multiple scales

repetitive patterns

time

consistency over

Motion

$$
\begin{aligned}
\min_{\Theta} \mathcal{L}(\Theta) = & \| \mathcal{H}^{\perp} \circ (\mathcal{X} - \text{Fold}(UV^T) - \mathcal{R}) \|_F^2 \\
& + \| \mathcal{H} \circ (\mathcal{X} - \mathcal{F} - \mathcal{R}) \|_F^2 + \lambda \| \mathcal{F} \|_{TV} \\
& + \alpha \| \mathcal{H} \|_{3DTV} + \beta \| \mathcal{H} \|_1 + b \sum_{k=1}^{K} \sum_{s=1}^{n_k} \| \mathcal{M}_{ks} \|_1 \\
s.t. \quad & \mathcal{R} = \sum_{k=1}^{K} \sum_{s=1}^{n_k} D_{ks} \otimes \mathcal{M}_{ks}, \quad \| D_{ks} \|_F^2 \leq 1.
\end{aligned} \tag{6}
$$

objective function (non-convex optimization):

Integrating these three models yields ***the final***

Moving objects layer, relative to

rain streak, **is more smooth**.

Regularization terms to

constrain the parameter space:

sparsity of **moving objects**

Spatial-temporal continuity and

Sparsity of rain positions

Alternating direction method of multipliers (ADMM) to iteratively optimize each variable.

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Video Decomposition